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هم نوا

Hamnawa

Accessible Document Reader for Visually Impaired Students

Functional and Non-Functional Requirements

Subject: Software Engineering **Group No.:** 2

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At a glance. This document defines **47 functional** and **28 non-functional** requirements for Hamnawa. Every requirement has a unique ID, a priority, a verification method and, for non-functional requirements, a measurable fit criterion, so that each one can be tested.

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1 Introduction

1.1 Purpose

This document lists what Hamnawa must *do* (functional requirements) and how *well* it must do it (non-functional requirements). It is the reference for design, implementation and testing. Hamnawa turns ordinary text-based PDF and PowerPoint documents into structured, spoken and navigable content for visually impaired and low-vision students.

1.2 Conventions

- **Wording.** "Shall" marks a mandatory statement.
- **IDs.** Functional requirements are `FR-nn`. Non-functional requirements are `NFR-xn`, where the letter is the quality group (U usability, A accessibility, P performance, R reliability, S security and privacy, C compatibility, M maintainability, L legal). IDs are never reused.
- **Priority (Pri.).** **M** = Must have, **S** = Should have, **C** = Could have.
- **Verification (Ver.).** **T** = Test, **D** = Demonstration, **I** = Inspection or review, **A** = Analysis.
- **Objective (Obj.).** The project objective that the requirement supports:

O1	Extract readable, correctly ordered text from PDF and PPTX files
O2	Recover document structure (headings, lists, slides) as navigable sections
O3	Read aloud with adjustable speed, voice, pause, resume and repeat
O4	Complete use without a mouse (keyboard and voice)
O5	Meet accessibility guidance (WCAG 2.2 AA) and work with screen readers

1.3 Summary of Functional Requirements

Functional group	Must	Should	Could	Total
Document Management	4	2	1	7
Extraction and Structure Recovery	7	3	0	10
Playback	5	4	1	10
Navigation	4	2	1	7
Voice Control	0	5	0	5
Bookmarks and Resume	0	4	0	4
Settings, Display and Help	2	2	0	4
Total	**22**	**22**	**3**	**47**

2 Functional Requirements

Each requirement can be tested on its own. Requirements are grouped by the part of the system they describe.

2.1 Document Management

How a document enters the system and how the user is told about it.

ID	Requirement	Pri.	Ver.	Obj.
FR-01	The system shall let the user select a PDF or PPTX file using a keyboard-operable file selector.	M	T	O4
FR-02	The system shall validate file type, file size (maximum 20 MB) and length (maximum 100 pages or slides) before processing, and report the result by speech and text.	M	T	O1
FR-03	The system shall detect files it cannot read (scanned or image-only, password-protected, corrupted, unsupported type) and state the specific reason and a suggested action.	M	T	O1
FR-04	The system shall show processing progress and announce when the document is ready.	S	D	O5
FR-05	When a document opens, the system shall announce its title (if found), number of pages or slides, and number of top-level sections.	M	D	O2

ID	Requirement	Pri.	Ver.	Obj.
FR-06	The system shall list recently opened documents (names only, stored locally) and let the user remove entries.	C	D	O4
FR-07	The system shall let the user erase all locally stored data (bookmarks, positions, settings, history) with one confirmed action.	S	T	O5

2.2 Extraction and Structure Recovery

Turning an ordinary file into a correctly ordered, structured document model.

ID	Requirement	Pri.	Ver.	Obj.
FR-08	The system shall extract text from text-based PDF files in correct reading order.	M	T	O1
FR-09	The system shall extract text from PPTX files: slide title, body text in order, and (user-selectable) speaker notes.	M	T	O1
FR-10	The system shall read multi-column page layouts column by column instead of line by line across columns.	S	T	O1
FR-11	The system shall detect and exclude repeated headers, footers and page numbers from the spoken text.	M	T	O1
FR-12	The system shall identify headings and their levels using font size, weight and numbering patterns.	M	T	O2
FR-13	The system shall identify paragraphs and bulleted or numbered lists, and announce lists with their item count.	M	T	O2
FR-14	The system shall identify slide boundaries and slide titles; a slide without a title shall be named “Slide N”.	M	T	O2
FR-15	The system shall build a hierarchical outline (sections, subsections, slides) that serves as the navigation model.	M	T	O2
FR-16	The system shall detect images, tables and charts and announce their presence and location (for example “Image on slide 7”) without describing their content.	S	T	O2

ID	Requirement	Pri.	Ver.	Obj.
FR-17	The system shall repair common extraction artefacts: hyphenated line breaks, ligatures and stray line breaks inside paragraphs.	S	T	O1

2.3 Playback

Speaking the document and giving the user control of the speech.

ID	Requirement	Pri.	Ver.	Obj.
FR-18	The system shall read the document aloud from the current position using text-to-speech.	M	T	O3
FR-19	The system shall provide pause, resume and stop of speech.	M	T	O3
FR-20	The system shall repeat the current sentence or paragraph on request.	M	T	O3
FR-21	The system shall skip to the next or previous sentence or paragraph on request.	M	T	O3
FR-22	The system shall let the user set speech rate between 0.5× and 3.0× in steps of 0.25×, taking effect within 1 second.	M	T	O3
FR-23	The system shall let the user choose among the voices installed on the device and adjust volume.	S	D	O3
FR-24	The system shall announce structural cues (heading level and text, list start, slide number) before reading the element, with an option to switch cues off.	S	T	O2
FR-25	The system shall provide a “continuous reading” setting that continues into the next section or stops at the end of each section, and shall announce the end of the document.	S	T	O3
FR-26	The system shall visually highlight the sentence currently being read.	C	D	O5
FR-27	The system shall offer a screen-reader coexistence mode in which the current text is exposed through a live region so the user’s own screen reader can read it instead of the built-in speech.	S	T	O5

2.4 Navigation

Moving around a document by structure, not just top to bottom.

ID	Requirement	Pri.	Ver.	Obj.
FR-28	The system shall move to the next or previous section on request.	M	T	O4
FR-29	The system shall move to the next or previous heading of a chosen level on request.	S	T	O2
FR-30	The system shall go directly to a given slide or page number.	M	T	O4
FR-31	The system shall provide an outline list that can be browsed with the arrow keys, with Enter jumping to the selected entry.	M	T	O4
FR-32	The system shall announce the current position (section title, slide or page, percentage read) on request.	S	T	O4
FR-33	The system shall jump to the beginning or end of the document on request.	M	T	O4
FR-34	The system shall search for a word or phrase and jump to its next occurrence.	C	T	O4

2.5 Voice Control

Hands-free operation using a small, fixed set of spoken commands.

ID	Requirement	Pri.	Ver.	Obj.
FR-35	The system shall recognise the English voice commands listed in Section 2.9 and perform the matching action.	S	T	O4
FR-36	The microphone shall be off by default and shall be switched on and off by the user with a keyboard shortcut or a spoken command.	S	T	O4
FR-37	The system shall give a short audible confirmation for a recognised command and a distinct signal for an unrecognised one.	S	D	O4
FR-38	The system shall pause or lower its own speech while listening for a command so that it does not hear itself.	S	T	O4
FR-39	If speech recognition is unavailable or permission is denied, the system shall say so and continue to work fully by keyboard.	S	T	O4

2.6 Bookmarks and Resume

Remembering where the user was and what they marked.

ID	Requirement	Pri.	Ver.	Obj.
FR-40	The system shall add a bookmark at the current position, with an optional typed or spoken label.	S	T	O4
FR-41	The system shall list bookmarks of the open document, jump to one, and delete one.	S	T	O4
FR-42	The system shall automatically save the last reading position of each document.	S	T	O4
FR-43	When the same document is opened again (identified by a content fingerprint), the system shall offer to resume from the saved position.	S	T	O4

2.7 Settings, Display and Help

Making the interface comfortable for low-vision users and easy to learn.

ID	Requirement	Pri.	Ver.	Obj.
FR-44	The system shall offer a high-contrast display mode and text size adjustable from 100% to 200%.	S	T	O5
FR-45	The system shall save user settings between sessions on the user's device.	S	T	O5
FR-46	The system shall provide a help command that reads out the available keyboard shortcuts and voice commands.	M	T	O5
FR-47	Every status and error message shall be delivered both as speech and as on-screen text.	M	I	O5

2.8 Keyboard Shortcuts (Proposed)

Single-key shortcuts work when the reader region has focus, as in media players. They are *proposed* and will be finalised after conflict testing with the NVDA screen reader and the browser (NFR-U4).

Key	Action	Requirement
`Space`	Play / pause	FR-18, FR-19
`Esc`	Stop	FR-19
`R`	Repeat current sentence	FR-20
← / →	Previous / next sentence	FR-21

Key	Action	Requirement
↑ / ↓	Previous / next paragraph	FR-21
`P` / `N`	Previous / next section	FR-28
`G`	Go to slide or page number	FR-30
`O`	Open outline	FR-31
`W`	Where am I	FR-32
`Home` / `End`	Start / end of document	FR-33
`+` / `-`	Faster / slower speech	FR-22
`B` / `L`	Add bookmark / list bookmarks	FR-40, FR-41
`M`	Microphone on / off	FR-36
`H` or `?`	Help	FR-46

2.9 Voice Commands

Spoken command	Action	Requirement
“play” / “resume”	Start or resume reading	FR-18, FR-19
“pause” / “stop”	Pause or stop reading	FR-19
“repeat”	Repeat the current sentence	FR-20
“next paragraph” / “previous paragraph”	Move by paragraph	FR-21
“next section” / “previous section”	Move by section	FR-28
“go to slide *n*” / “go to page *n*”	Jump to a slide or page	FR-30
“outline”	Open the outline	FR-31
“where am I”	Announce position	FR-32
“faster” / “slower”	Change speech rate	FR-22
“bookmark” / “list bookmarks”	Add or list bookmarks	FR-40, FR-41
“help”	Read the help	FR-46
“microphone off”	Switch voice control off	FR-36

3 Non-Functional Requirements

Each non-functional requirement has a *fit criterion*: a measurable statement that tells the tester when it is met. Numeric targets marked "target" are first estimates and will be confirmed by measurement in the first increment.

3.1 Usability

ID	Requirement	Fit criterion (measure)	Ver.
NFR-U1	A new user shall be able to open a document and start listening using the keyboard only.	At most 3 steps from page load to speech starting.	T
NFR-U2	Every function shall be reachable without a mouse.	100% of functional requirements operable by keyboard; checked by a keyboard-only walkthrough.	T
NFR-U3	The product shall be learnable without training.	At least 80% of test users complete the core task (open, listen, navigate, bookmark) within 10 minutes without help.	T
NFR-U4	Shortcuts shall be consistent and shall not clash with common screen-reader or browser keys.	No conflicts found in a documented shortcut review with NVDA and the browser.	I

3.2 Accessibility

ID	Requirement	Fit criterion (measure)	Ver.
NFR-A1	The interface shall conform to WCAG 2.2 level AA principles.	Zero open failures of AA criteria in the final accessibility checklist.	I
NFR-A2	All controls shall have text labels, a logical focus order and a visible focus indicator.	100% of controls pass automated and manual checks.	T
NFR-A3	Text and interface elements shall have sufficient contrast.	Contrast ratio of at least 4.5:1 for text, 3:1 for controls.	A
NFR-A4	No information shall be conveyed by colour alone.	Checked by inspection of every status indicator.	I
NFR-A5	The product shall work with the NVDA screen reader.	All core tasks completed with NVDA running.	T

ID	Requirement	Fit criterion (measure)	Ver.
NFR-A6	The layout shall remain usable when zoomed.	No loss of content or function at 200% browser zoom.	T

3.3 Performance

ID	Requirement	Fit criterion (measure)	Ver.
NFR-P1	Reading shall begin promptly after a document is opened.	Speech starts within 5 s for a 50-page document (target, to be validated in Increment 1).	T
NFR-P2	Keyboard commands shall feel immediate.	Response (speech or focus change) within 200 ms for 95% of key presses.	T
NFR-P3	Voice commands shall be acted on quickly.	Within 2 s of the end of the spoken command.	T
NFR-P4	Document processing shall complete in reasonable time.	At most 20 s for a 100-page document on a typical student laptop (target).	T

3.4 Reliability

ID	Requirement	Fit criterion (measure)	Ver.
NFR-R1	A bad or unsupported file shall never crash the application.	All files in the negative test set produce a spoken error, with no crash.	T
NFR-R2	Bookmarks and positions shall not be lost on reload or browser restart.	100% of saved items restored in the persistence test.	T
NFR-R3	If speech output fails, the user shall be told and the text remain available.	Failure simulated; message shown and text readable.	T

3.5 Security and Privacy

ID	Requirement	Fit criterion (measure)	Ver.
NFR-S1	Documents shall be processed on the user's device where technically possible; any server copy shall be deleted at session end.	No document stored after the session; verified by design review and test.	I
NFR-S2	Bookmarks, positions and settings shall be stored locally only.	No personal data sent to the project's servers.	I
NFR-S3	The user shall be told that browser speech recognition may send audio to the browser vendor.	Notice shown before the microphone is first enabled.	I
NFR-S4	File parsing shall be defensive against malformed files.	Fuzzed and malformed samples do not cause crashes or hangs.	T

3.6 Compatibility

ID	Requirement	Fit criterion (measure)	Ver.
NFR-C1	The product shall run in current desktop Chrome and Edge on Windows 10 and 11.	Core tasks pass on the latest two versions of each.	T
NFR-C2	The product shall remain usable on small laptop screens.	Fully usable at 1366×768.	T
NFR-C3	Firefox shall be supported on a best-effort basis for keyboard and speech output.	Voice commands may be unavailable; stated to the user.	D

3.7 Maintainability

ID	Requirement	Fit criterion (measure)	Ver.
NFR-M1	The code shall be organised in modules: extraction, structuring, playback, navigation, interface.	Each module has a documented interface and can be tested alone.	I

ID	Requirement	Fit criterion (measure)	Ver.
NFR-M2	Core logic shall be covered by automated tests.	At least 70% statement coverage for extraction and structuring modules.	A
NFR-M3	The code shall follow an agreed coding standard and be reviewed.	Every merge to the main branch reviewed by a second member.	I

3.8 Legal and Licensing

ID	Requirement	Fit criterion (measure)	Ver.
NFR-L1	All third-party libraries and fonts shall have licences compatible with academic use and redistribution.	Licence list reviewed and recorded in the design document.	I

4 Traceability

4.1 Objectives to Functional Requirements

Obj.	Objective	Functional requirements
O1	Extract readable, correctly ordered text	FR-02, FR-03, FR-08, FR-09, FR-10, FR-11, FR-17
O2	Recover document structure	FR-05, FR-12, FR-13, FR-14, FR-15, FR-16, FR-24, FR-29
O3	Read aloud with adjustable controls	FR-18, FR-19, FR-20, FR-21, FR-22, FR-23, FR-25
O4	Complete use without a mouse (keyboard and voice)	FR-01, FR-06, FR-28, FR-30, FR-31, FR-32, FR-33, FR-34, FR-35, FR-36, FR-37, FR-38, FR-39, FR-40, FR-41, FR-42, FR-43
O5	Accessibility and screen-reader compatibility	FR-04, FR-07, FR-26, FR-27, FR-44, FR-45, FR-46, FR-47

Non-functional requirements apply across all objectives; accessibility requirements (NFR-A1 to NFR-A6) support O5 directly.

4.2 Requirements to Tests

Every requirement whose verification method is T shall have at least one test case in the Test Plan, named after the requirement ID (for example `TC-FR-22-01`). The traceability matrix will be kept in the repository and updated at the end of every increment.

5 Open Issues and Validation

These requirements are derived from the project proposal, accessibility guidance and the team's analysis. They have *not yet* been validated with real users; interviews with visually impaired or low-vision students and a formal team review are planned.

ID	Issue	Owner / when
OI-1	Confirm the 20 MB and 100-page limits (FR-02) against typical course files.	Team, Increment 1
OI-2	Finalise keyboard shortcuts after conflict testing with NVDA and the browser.	[owner], Increment 2
OI-3	Confirm that the performance targets (NFR-P1 to P4) are realistic.	[owner], Increment 1
OI-4	Interview at least one visually impaired or low-vision student and update priorities.	[owner], [date]

A Glossary

Screen reader Software that reads on-screen content aloud for blind or low-vision users (for example NVDA).

Text-to-speech (TTS) Technology that turns written text into spoken audio.

Speech recognition Technology that turns spoken words into text or commands.

WCAG Web Content Accessibility Guidelines, published by the W3C.

OCR Optical character recognition; reads text from images.

Outline The hierarchical list of sections, subsections and slides of a document.

Fingerprint A short value computed from a file's content that identifies it.

Live region A part of a web page whose changes are announced by screen readers automatically.

MoSCoW Prioritisation: Must, Should, Could, Won't.

Fit criterion A measurable statement that defines when a requirement is satisfied.